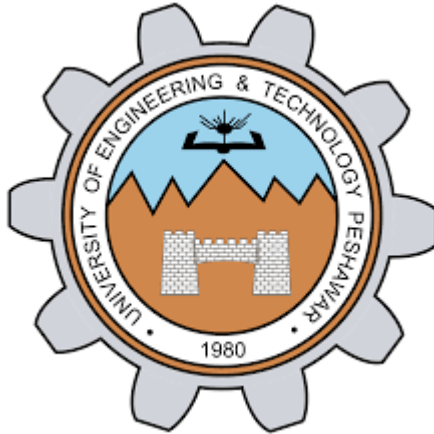




Lab Report – EE-170L Computer Programming (Lab 11)

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Objectives

- To understand C-Strings (character arrays terminated with `\0`).
- To learn declaration and initialization of C-Strings.
- To explore String objects in C++.
- To apply string functions for manipulation (length, concatenation, comparison, etc.).

Theory

1. C-Strings

- A C-String is an array of characters ending with a null character `\0`.

Example:

```
char myString[] = "Hello";
```

2.String Objects

- The `string` class in C++ (from `<string>` header) allows dynamic length strings.
- Input can be taken using `getline(cin, str)` which reads entire lines including spaces.

3. String Functions

Common C-String functions include:

- `strlen(s)` → length of string
- `strcpy(s1, s2)` → copy string
- `strcat(s1, s2)` → concatenate strings
- `strcmp(s1, s2)` → compare strings
- `strrev(s)` → reverse string
- `strlwr(s)` → convert to lowercase
- `strupr(s)` → convert to uppercase

Lab Tasks

Task 1: String Length Calculation

```
#include <iostream>    // tells the compiler to include input output stream library
#include <cstring>      // include string functions
using namespace std;   // tells the compiler to use standard namespace

int main() {           // main function begins from here
    char message[] = "Electrical Engineering"; // declare and initialize string
    cout << "Message: " << message << endl;    // display the string
    cout << "Length of message: " << strlen(message) << endl; // calculate and
    show length
    return 0;          // end program
}
```

Task 2: String Concatenation

```
#include <iostream>    // include input output stream library
#include <cstring>      // include string functions
using namespace std;   // use standard namespace std

int main() {           // main function starting point
    char firstName[20] = "Naeem"; // first name
    char lastName[20] = " ul Haq"; // last name
    strcat(firstName, " ");        // add space after first name
```

```
    strcat(firstName, lastName);           // join last name to first name

    cout << "Full Name: " << firstName << endl; // display full name

    return 0;                             // end program
}
```

Conclusion

In this lab, we explored both C-Strings and String objects in C++. C-Strings are fixed-size character arrays terminated with a null character, while String objects provide more flexibility with dynamic length and easier handling of spaces. We practiced using built-in functions such as `strlen()` for measuring string length and `strcat()` for concatenation, which demonstrated how text data can be manipulated efficiently.

The exercises showed the limitations of `cin >>` (stopping at spaces) and the advantages of `cin.get()` and `getline()` for reading complete lines. Overall, this lab strengthened our understanding of how strings are stored in memory, how they can be processed, and why modern C++ string objects are often preferred for real-world applications.